

Intermediate Dynamics and Vibrations | MAE 5730

HW Problem 30 | “Cart and pendulum”

Aditya Bhaskar | ab2823

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1 THE PROBLEM STATEMENT

Cart m_1 slides on a frictionless surface. A simple pendulum of mass m_2 and length l is attached to it. $\theta = 0$ corresponds to the stable equilibrium position. Use gravity g . Find the governing equations using:

(a) The Newton-Euler equations.

(b) The Lagrange equations.

Show that both methods give the same result.

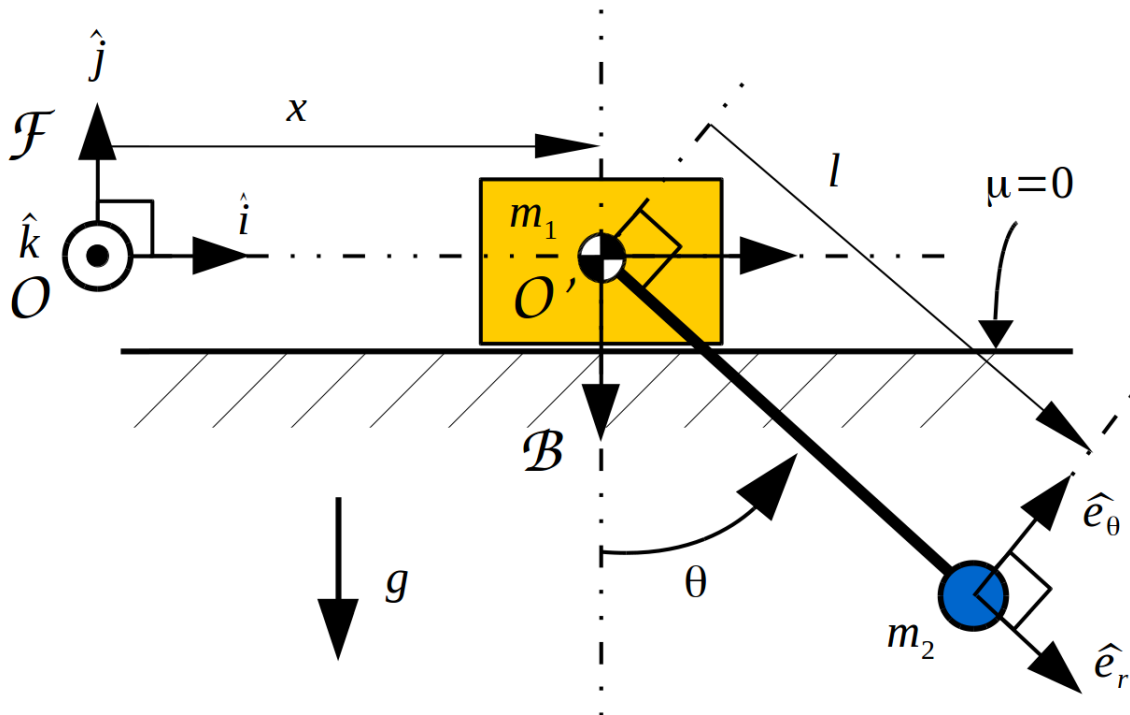


Figure 1.1: The cart-pendulum system

Assumptions:

- The surface on which the cart moves is horizontal and frictionless.
- The connecting-rod is rigid and massless, and passes through the centres of mass of both the cart and the particle.
- The radius of the particle is infinitesimal.
- The gravitational field is uniform and acts vertically downwards.

Reference frames:

- Frame $\mathcal{F}$, with origin O , is fixed in space. Cartesian coordinates are used in frame $\mathcal{F}$.
- Frame $\mathcal{B}$, with origin O' , is attached to the moving cart. Polar coordinates are used in frame $\mathcal{B}$.

Fixed parameters:

- m_1 = Mass of the cart
- m_2 = Mass of the particle
- l = Length of the rod
- g = Acceleration due to gravity

Variables:

- θ = Angular displacement of the rod from the vertical (counter-clockwise positive)
- x = Horizontal displacement of the cart, measured in frame $\mathcal{F}$

Other symbols:

- N = Normal reaction on the cart due to the surface
- T = Tension force in the rod
- y = Vertical displacement in frame $\mathcal{F}$
- $\vec{r}$ = Position vector
- $\vec{v}$ = Velocity vector
- $\vec{a}$ = Acceleration vector
- $\vec{\alpha}$ = Angular acceleration of a rigid body
- $\vec{F}$ = Force vector
- $\hat{i}, \hat{j}, \hat{k}$ = Cartesian unit vectors in frame $\mathcal{F}$
- $\hat{e}_r, \hat{e}_\theta, \hat{k}$ = Polar unit vectors in frame $\mathcal{B}$
- E_P = Potential energy
- E_K = Kinetic energy
- $\mathcal{L}$ = Lagrangian
- μ = Coefficient of friction between the surface and the cart
- M = Total mass of a rigid body
- I_G = Moment of inertia of a rigid body about its center of mass, G

2 FREE-BODY DIAGRAMS (FBDs)

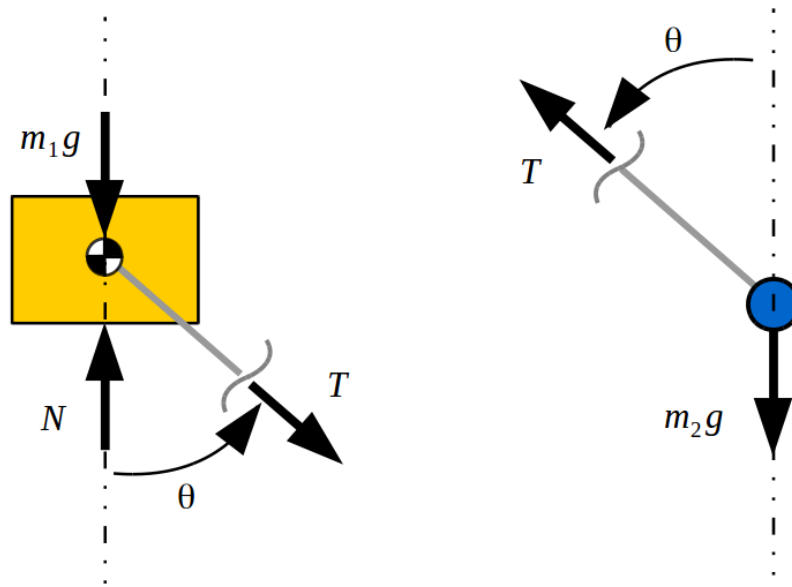


Figure 2.1: Free-Body diagrams of the cart and the particle in an arbitrary configuration

Aside:

In Fig. 2.1, it is important to note that the rod does not transmit any moment, and the forces exerted by the rod are along the rod. This is deduced by considering the Free-Body Diagram of the rod-particle system, as shown in Fig. 2.2. Since there is a pin joint at O' , it is incapable of exerting a moment on the rod at O' .

AMB about the particle:

$$\sum_{i, \text{external}} \vec{r}_{i/2} \times \vec{F}_i = I_G \vec{\alpha} + M(\vec{r}_{G/2} \times \vec{a}_G). \quad (2.1)$$

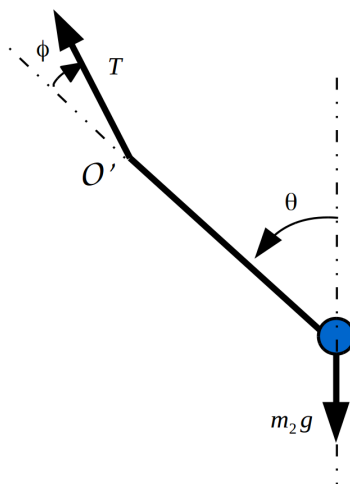


Figure 2.2: Free-Body diagram of the rod-particle system

Since the rod is massless, the center of mass of this system, G , coincides with the particle. Furthermore since the radius of the particle is infinitesimal, the moment of inertia of this system about the center of mass of the system (equivalently about the particle), I_G , is zero. Thus we can write Eqn (2.1) as,

$$\begin{aligned} -l\hat{e}_r \times \vec{T} + \vec{0} \times (-m_2 g \hat{j}) &= 0(\vec{\alpha}) + M(\vec{0} \times \vec{a}_G), \\ \Rightarrow \vec{T} \times \hat{e}_r &= \vec{0}, \\ \Rightarrow \vec{T} &= T\hat{e}_r, \end{aligned} \quad (2.2)$$

where T can be either positive or negative. Since the rod is massless this is the force acted upon the cart and the particle due to the rod, $\vec{T}$, and it acts along rod. This is equivalent to saying $\vec{T} = T\hat{e}_r$ or the angle ϕ in Fig. 2.2 is 0 or π radians.

3 "CART AND PENDULUM" SOLVED USING THE NEWTON-EULER EQUATIONS

LMB in the x-direction:

$$m_1 a_{x,1} + m_2 a_{x,2} = \sum_{\text{external}} F_x.$$

Since the frictional force between the cart and the surface is zero, the net external forces on the system, in the x-direction, is also zero, and we write,

$$\Rightarrow m_1 a_{x,1} + m_2 a_{x,2} = 0. \quad (3.1)$$

The linear momentum of the cart-pendulum system is conserved in the x -direction. Now,

$$a_{x,1} = \ddot{x}, \quad (3.2)$$

and,

$$\begin{aligned} a_{x,2} &= \frac{d^2}{dt^2} (x + l \sin \theta), \\ \Rightarrow a_{x,2} &= \ddot{x} - l\dot{\theta}^2 \sin \theta + l\ddot{\theta} \cos \theta. \end{aligned} \quad (3.3)$$

Substituting Eqns (3.2) and (3.3) into Eqn (3.1), we have,

$$\Rightarrow \boxed{m_1 \ddot{x} + m_2 (\ddot{x} - l\dot{\theta}^2 \sin \theta + l\ddot{\theta} \cos \theta) = 0} \quad (3.4)$$

AMB about O':

$$\begin{aligned} \sum \vec{F}_2 &= m_2 \vec{a}_2, \\ \Rightarrow \vec{r}_{2/O'} \times \sum \vec{F}_2 &= \vec{r}_{2/O'} \times m_2 \vec{a}_2. \end{aligned} \quad (3.5)$$

Eqn (3.5) is valid even for accelerating O' . Also,

$$\begin{aligned} \vec{a}_2 &= \vec{a}_{2/O'} + \vec{a}_{O'/O}, \\ \Rightarrow \vec{a}_2 &= (-l\dot{\theta}^2 \hat{e}_r + l\ddot{\theta} \hat{e}_\theta) + \ddot{x} \hat{i}. \end{aligned} \quad (3.6)$$

Using the Free-Body Diagrams and substituting Eqn (3.6) into (3.5), we have,

$$\begin{aligned} \Rightarrow l\hat{e}_r \times (T\hat{e}_r - m_2 g \hat{j}) &= l\hat{e}_r \times m_2 (-l\dot{\theta}^2 \hat{e}_r + l\ddot{\theta} \hat{e}_\theta + \ddot{x} \hat{i}), \\ \Rightarrow -m_2 g l \sin \theta \hat{k} &= m_2 (l^2 \ddot{\theta} \hat{k} + \ddot{x} l \cos \theta \hat{k}), \\ \Rightarrow -m_2 g l \sin \theta &= m_2 (l^2 \ddot{\theta} + \ddot{x} l \cos \theta), \\ \Rightarrow \boxed{g \sin \theta + l\ddot{\theta} + \ddot{x} \cos \theta} &= 0 \end{aligned} \quad (3.7)$$

Notice that we have not written the RHS of Eqn (3.5) as a derivative of the angular momentum of the particle, but have instead used the cross product of the equation with a suitable position vector to "kill" the term, T , that is not of interest.

Eqns (3.4) and (3.7) are the required non-linear governing equations describing the cart-pendulum system, derived using the Newton-Euler equations.

4 “CART AND PENDULUM” SOLVED USING THE LAGRANGE EQUATIONS

The Lagrangian:

$$\mathcal{L} = E_K - E_P. \quad (4.1)$$

Taking the gravitational potential energy to be zero at the reference level $y = 0$, we write,

$$\begin{aligned} E_P &= E_{P1} + E_{P2}, \\ \Rightarrow E_P &= -m_2 g l \cos \theta. \end{aligned} \quad (4.2)$$

Also,

$$\begin{aligned} E_K &= E_{K,1} + E_{K,2}, \\ \Rightarrow E_K &= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2. \end{aligned} \quad (4.3)$$

But,

$$v_1^2 = \dot{x}^2. \quad (4.4)$$

And,

$$\begin{aligned} v_2^2 &= (\dot{x} + l\dot{\theta} \cos \theta)^2 + (l\dot{\theta} \sin \theta)^2, \\ \Rightarrow v_2^2 &= \dot{x}^2 + l^2 \dot{\theta}^2 + 2\dot{x}l\dot{\theta} \cos \theta. \end{aligned} \quad (4.5)$$

Substituting Eqns (4.4) and (4.5) into (4.3), we have,

$$\Rightarrow E_K = \frac{1}{2} m_1 \dot{x}^2 + \frac{1}{2} m_2 (\dot{x}^2 + l^2 \dot{\theta}^2 + 2\dot{x}l\dot{\theta} \cos \theta). \quad (4.6)$$

Now using Eqns (4.2) and (4.6), we write the expression for the Lagrangian, $\mathcal{L}$, as,

$$\Rightarrow \mathcal{L} = \frac{1}{2} m_1 \dot{x}^2 + \frac{1}{2} m_2 (\dot{x}^2 + l^2 \dot{\theta}^2 + 2\dot{x}l\dot{\theta} \cos \theta) + m_2 g l \cos \theta. \quad (4.7)$$

Lagrange equation for x :

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) &= 0, \\ \Rightarrow 0 - \frac{d}{dt} ((m_1 + m_2) \dot{x} + m_2 l \dot{\theta} \cos \theta) &= 0, \\ \Rightarrow \boxed{m_1 \ddot{x} + m_2 (\ddot{x} - l \dot{\theta}^2 \sin \theta + l \ddot{\theta} \cos \theta)} &= 0 \end{aligned} \quad (4.8)$$

Lagrange equation for θ :

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) &= 0, \\ \Rightarrow -m_2 \dot{x} l \dot{\theta} \sin \theta - m_2 g l \sin \theta - \frac{d}{dt} (m_2 l^2 \dot{\theta} + m_2 \dot{x} l \cos \theta) &= 0, \\ \Rightarrow -m_2 \dot{x} l \dot{\theta} \sin \theta - m_2 g l \sin \theta - m_2 l^2 \ddot{\theta} - m_2 \ddot{x} l \cos \theta + m_2 \dot{x} l \dot{\theta} \sin \theta &= 0, \\ \Rightarrow \boxed{g \sin \theta + l \ddot{\theta} + \ddot{x} \cos \theta} &= 0 \end{aligned} \quad (4.9)$$

Eqns (4.8) and (4.9) are the required non-linear governing equations describing the cart-pendulum system, derived using the Lagrange equations.

We note that Eqns (3.4) and (3.7) are identical to (4.8) and (4.9), respectively. Thus we have conclusively shown that the governing equations derived using the Newton-Euler equations and the Lagrange equations are the same.